

NEW DELHI SAFDARJUNG, India

WMO: 421820

Lat: 28.5886N

Lon: 77.2222E

Elev: 215

StdP: 98.77

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.0	(c) 7.1	(d) 0.5	(e) 4.0	(f) 20.9	(g) 2.4	(h) 4.6	(i) 19.6	(j) 7.3	(k) 19.3	(l) 6.3	(m) 18.4	(n) 0.5	(o) 270	(p) 0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 9.7	(c) 42.3	(d) 23.2	(e) 41.0	(f) 23.6	(g) 39.6	(h) 23.7	(i) 28.9	(j) 34.6	(k) 28.6	(l) 33.9	(m) 28.2	(n) 33.3	(o) 3.7	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 27.7	(b) 24.3	(c) 31.3	(d) 27.3	(e) 23.7	(f) 31.1	(g) 27.0	(h) 23.3	(i) 30.9	(j) 96.5	(k) 34.7	(l) 94.6	(m) 34.1	(n) 92.9	(o) 33.6	(p) 31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.0	(b) 5.9	(c) 5.0	(e) 4.1	(f) 44.4	(g) 1.5	(h) 1.0	(i) 3.1	(j) 45.1	(k) 2.2	(l) 45.6	(m) 1.4	(n) 46.2	(o) 0.3	(p) 46.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.5	14.1	17.8	23.4	29.5	32.9	33.3	31.4	30.4	29.6	26.4	20.8	15.7		
	DBStd	7.03	2.30	2.66	3.09	2.78	2.70	2.91	2.11	1.62	1.74	2.23	2.17	2.49		
	HDD10.0	2	1	0	0	0	0	0	0	0	0	0	0	1		
	HDD18.3	264	133	38	2	0	0	0	0	0	0	0	4	87		
	CDD10.0	5651	127	219	414	584	711	700	663	633	589	508	325	178		
	CDD18.3	2872	1	24	158	334	453	450	404	374	339	250	79	6		
	CDH23.3	38596	13	207	1569	4512	7071	7086	5754	5009	4247	2451	621	56		
(13)	CDH26.7	22208	0	38	695	2722	4729	4736	3304	2575	2081	1177	149	2		
(14)	Wind	WSAvg	1.6	1.4	1.7	1.9	2.0	2.1	1.9	1.7	1.5	0.9	0.8	1.0		
(15) Precipitation	PrecAvg	780	17	22	17	10	24	78	211	246	131	18	4	11		
	PrecMax	1232	80	123	97	61	78	203	632	583	379	93	39	72		
	PrecMin	380	0	0	0	0	0	3	7	53	0	0	0	0		
	PrecStd	230	19	25	20	14	24	52	137	133	92	26	8	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	30.2	37.0	41.7	44.1	44.2	39.9	37.0	36.7	35.8	31.6	26.7		
		MCWB	16.9	19.1	20.3	21.7	22.2	23.2	26.4	27.2	25.4	23.6	19.6	17.1		
	2%	DB	22.9	27.6	34.4	40.0	42.6	42.5	38.1	35.9	35.7	34.6	29.8	24.9		
		MCWB	15.5	18.0	20.0	21.4	22.6	24.0	27.0	27.4	25.5	23.4	19.4	16.7		
	5%	DB	21.2	25.9	32.4	38.7	41.2	41.0	36.8	35.0	34.9	33.5	28.5	23.4		
		MCWB	14.2	16.8	19.5	21.0	22.8	25.1	27.3	27.4	25.6	23.0	18.8	15.6		
	10%	DB	19.7	24.1	30.7	37.1	39.9	39.4	35.5	34.0	34.0	32.3	27.1	21.9		
		MCWB	13.4	15.9	19.1	20.7	23.1	25.4	27.6	27.4	25.7	22.4	18.2	14.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	20.8	23.2	25.6	28.2	29.2	29.5	29.2	28.7	27.0	22.2	19.1		
		MCDB	23.2	26.4	32.0	37.3	37.9	36.6	34.4	33.8	33.4	32.7	28.0	24.3		
	2%	WB	16.6	19.2	21.8	24.0	27.0	28.6	29.0	28.7	28.0	25.9	21.1	17.7		
		MCDB	21.1	25.8	30.5	35.0	36.5	36.3	34.3	33.2	32.5	31.3	27.3	23.0		
	5%	WB	15.5	18.0	20.9	22.9	26.0	28.1	28.6	28.3	27.5	24.9	20.1	16.6		
		MCDB	19.3	23.9	29.0	34.2	35.9	35.4	33.7	32.7	31.8	30.0	25.9	21.7		
	10%	WB	14.5	17.0	20.0	22.1	25.2	27.6	28.2	28.0	26.9	24.1	19.3	15.6		
		MCDB	18.0	22.0	27.8	33.5	35.2	34.5	33.1	32.2	31.1	29.5	24.9	20.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.1	13.0	13.9	12.6	9.7	6.8	6.4	8.0	12.0	13.5	12.6		
		MCDBR	13.5	13.7	14.8	15.3	14.0	11.6	8.3	7.6	9.2	12.4	14.3	14.4		
		MCWBR	6.4	5.3	4.1	4.4	4.2	3.8	2.8	2.7	2.7	3.8	5.5	6.8		
	5% WB	MCDBR	11.6	12.5	13.6	13.4	11.7	9.0	7.1	6.7	7.4	9.9	12.7	13.2		
		MCWBR	6.5	5.5	4.7	4.9	4.8	3.6	2.8	2.7	2.9	3.7	5.6	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.511	0.487	0.502	0.577	0.675	0.745	0.751	0.656	0.584	0.662	0.636	0.574		
	taud		1.814	1.931	1.924	1.784	1.634	1.551	1.563	1.755	1.872	1.620	1.619	1.695		
	Ebn at Noon		696	771	791	746	676	627	621	678	713	610	579	609		
	Edh at Noon		182	175	187	222	259	280	276	226	195	237	221	198		
(44) All-Sky Solar Radiation	RadAvg		3.02	4.28	5.62	6.61	6.76	5.97	4.91	4.83	5.02	4.60	3.57	3.03		
	RadStd		0.37	0.35	0.26	0.21	0.30	0.45	0.43	0.41	0.45	0.22	0.23	0.23		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.29	N/A	+2.01	N/A	+0.38	+0.33	N/A	N/A	+102	+107
(47)	Regional (2 neighbors)	N/A	N/A	+1.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page